Unbounded Electromagnetic Waves

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Lecture Objective

- To demonstrate a uniform plane wave solution to Maxwell's equations.

Electromagnetic Waves

- A time-varying electric field gives rise to a time-varying magnetic field, and a time-varying magnetic field gives rise to a time-varying electric field.
- This interchange results in the propagation of electromagnetic waves away from the source at a finite velocity.

Spherical Waves (1 of 2)

- If the wave propagates through a homogeneous medium without interacting with any obstacles or change in the medium, then the wave is said to be unbounded.
- Imagine a star in an otherwise empty universe. The light from this star is an unbounded electromagnetic wave propagating in free space.
- Far away, the star looks like a point source.
- The electromagnetic waves emanating from a point source are spherical waves.

Spherical Waves (2 of 2)

- The field has the same phase everywhere on a sphere of given radius with the point source at its center.
- We say the field comprises spherical wavefronts.
- Far away, antennas also look like point sources with spherical waves emanating from their phase center. However, they usually radiate more energy in certain directions (that is, they have directivity).

Wavefront

- A <u>wavefront</u> is a surface of constant time delay relative to the source and perpendicular to the direction of wave propagation.
- For time-harmonic (sinusoidal) sources, a wavefront is a surface of constant phase.

Plane Waves

- Very far from the source, the spherical wavefronts appear planar and can be treated locally as uniform plane waves.
- A uniform plane wave has identical properties everywhere on its planar wavefront.

Simple Medium

- The constitutive parameters (permittivity, permeability, and conductivity) of a simple medium are:
  - Linear
  - Time-invariant
  - Homogeneous
  - Isotropic

Lossless and Lossy Media

- Waves propagating in a lossless medium do not lose energy to the medium; the same power is spread over every spherical wavefront regardless of distance from the source.
- Waves propagating in a lossy medium lose energy to the medium in the form of heat; the wave power attenuates as it propagates.

Free space is a simple and lossless medium

Guided Waves

- When wave propagation is confined by material boundaries, the wave is said to be guided.
- Guided waves exist in transmission lines, hollow tube waveguides, etc.
- Earth's surface and ionosphere form a waveguiding structure that allows long distance communication in the HF range (3 – 30 MHz) and for lower frequencies.
- Lower frequency waves tend to follow the earth's curvature (ground waves).
- Electromagnetic waves trapped inside printed circuit boards (substrate modes) can be problematic causing unwanted coupling and interference.

Electromagnetic Wave Equation from Maxwell's Equations (1 of 3)

General form of Maxwell's equations:

$$\nabla \times \overline{E} = -\frac{\partial \overline{B}}{\partial t}$$

$$\nabla \times \overline{H} = \overline{J}_i + \overline{J}_c + \frac{\partial \overline{D}}{\partial t}$$

$$\nabla \cdot \overline{D} = \rho$$

$$\nabla \cdot \overline{B} = 0$$

In a source-free region of free space:

$$\nabla \times \overline{E} = -\mu_0 \frac{\partial \overline{H}}{\partial t} \quad (1)$$

$$\nabla \times \overline{H} = \varepsilon_0 \frac{\partial \overline{E}}{\partial t} \quad (2)$$

$$\nabla \cdot \overline{E} = 0 \quad (3)$$

$$\nabla \cdot \overline{H} = 0 \quad (4)$$

Electromagnetic Wave Equation from Maxwell's Equations (2 of 3)

$\nabla \times (1)$:

Using (2)

$$\nabla \times \nabla \times \overline{E} = -\mu_0 \frac{\partial}{\partial t}\left(\nabla \times \overline{H}\right) = -\varepsilon_0 \mu_0 \frac{\partial^2 \overline{E}}{\partial t^2}$$

Vector identity:

Using (4)

$$\nabla \times \nabla \times \overline{E} = \nabla\left(\nabla \bullet \overline{E}\right) - \nabla^2 \overline{E} = -\nabla^2 \overline{E}$$

Wave Equation for Electric Field

$$\boxed{\nabla^2 \overline{E} - \varepsilon_0 \mu_0 \frac{\partial^2 \overline{E}}{\partial t^2} = 0}$$

Electromagnetic Wave Equation from Maxwell's Equations (3 of 3)

$\nabla \times (2)$:

Using (1)

$$\nabla \times \nabla \times \overline{H} = \varepsilon_0 \frac{\partial}{\partial t}\left(\nabla \times \overline{E}\right) = -\varepsilon_0 \mu_0 \frac{\partial^2 \overline{H}}{\partial t^2}$$

Vector identity:

Using (3)

$$\nabla \times \nabla \times \overline{H} = \nabla\left(\nabla \bullet \overline{H}\right) - \nabla^2 \overline{H} = -\nabla^2 \overline{H}$$

Wave Equation for Magnetic Field

$$\boxed{\nabla^2 \overline{H} - \varepsilon_0 \mu_0 \frac{\partial^2 \overline{H}}{\partial t^2} = 0}$$

Electromagnetic Wave Equations in Free Space from Maxwell's Equations

Wave Equation for Electric Field

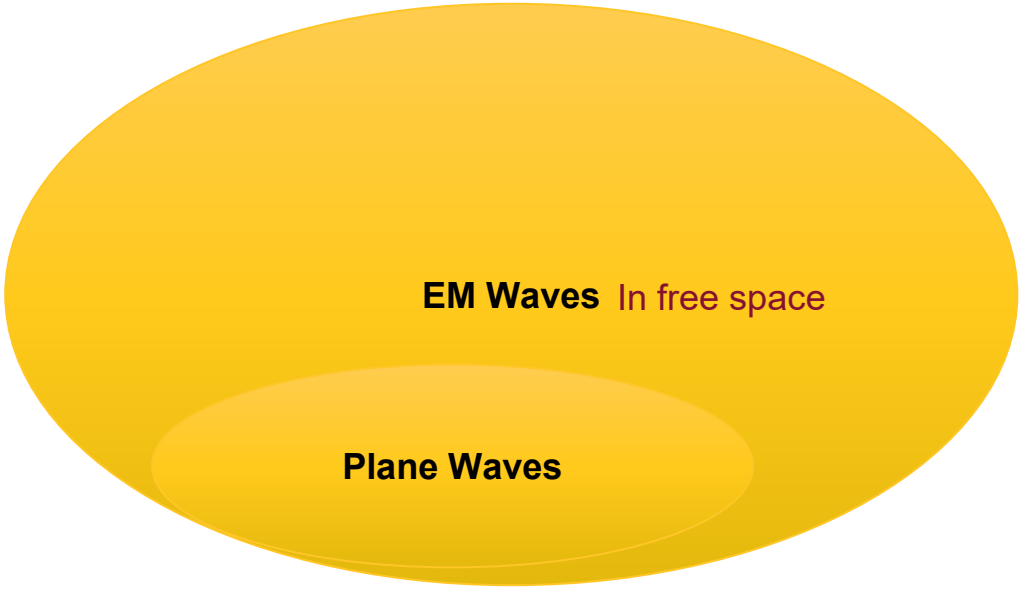
$$\nabla^2 \bar{E} - \frac{1}{c^2}\frac{\partial^2 \bar{E}}{\partial t^2} = 0$$

Wave Equation for Magnetic Field

$$\nabla^2 \bar{H} - \frac{1}{c^2}\frac{\partial^2 \bar{H}}{\partial t^2} = 0$$

We only need to solve one of these equations since E and H are related by Maxwell's equations …

Wave Solutions to Maxwell's Eqs



Plane Wave Solution

- The simplest solutions to the electromagnetic wave equations are plane waves.
- Let's assume a uniform plane wave solution propagating in the z-direction with only an x-directed component of the electric field:

$$\bar{E} = \hat{x}E_x(z,t)$$

Establishment of this specific form of plane wave solution requires a specific form for the source.

Uniform Plane Waves

- The wavefronts are planes perpendicular to the direction of propagation.
- The electric and magnetic fields are constant over a given wavefront.

Plane Wave Solution (1 of 3)

- Under these conditions:

$$\frac{\partial}{\partial x} \equiv 0$$
$$\frac{\partial}{\partial y} \equiv 0$$

$$\Rightarrow \quad \nabla^2 \bar{E} = \hat{a}_x \frac{\partial^2 E_x}{\partial z^2}$$

Plane Wave Solution (2 of 3)

- **Wave equation for electric field simplifies to:**

$$\frac{\partial^2 E_x}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2 E_x}{\partial t^2} = 0$$

1-D wave equation

Plane Wave Solution (3 of 3)

- Solutions to the above DE can be written as

$$E_x(z,t) = f\left(t \mp \frac{z}{c}\right)$$

- These solutions are waves.
- A wave is a disturbance that propagates through space as a function of time with a certain velocity that usually depends on the properties of the medium.

Checking …

$$\frac{\partial^2 f\left(t \mp \frac{z}{c}\right)}{\partial z^2} - \frac{1}{c^2}\frac{\partial^2 f\left(t \mp \frac{z}{c}\right)}{\partial t^2} = 0$$

$$\frac{1}{c^2} f''\left(t \mp \frac{z}{c}\right) - \frac{1}{c^2} f''\left(t \mp \frac{z}{c}\right) = 0$$

Plane Wave Solution

- The precise function *f*(*) is determined by the source and the boundary conditions.
- The canonical source of a plane wave is an infinite current sheet with no variation in the current as a function of position on the sheet.
- The simplest boundary condition is the radiation condition – waves propagate outward from the source to ± infinity.

Direction of Propagation

$f(t - {}^{z}/_{c})$ → Wave propagating in +z-direction

$f(t + {}^{z}/_{c})$ → Wave propagating in -z-direction

Interconnection Between Electric and Magnetic Fields of Wave Solutions to Maxwell's Equations (1 of 2)

- A time-varying electric field cannot exist without a time-varying magnetic field, and vice versa.
- Given the electric field, the corresponding magnetic field can be found using Faraday's law.

$$-\frac{\partial \bar{B}}{\partial t} = \nabla \times \bar{E}$$

Interconnection Between Electric and Magnetic Fields of Wave Solutions to Maxwell's Equations (2 of 2)

- For the electric field solution obtained above, we have

$$-\frac{\partial \bar{H}}{\partial t} = \frac{1}{\mu_0}\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 0 & \frac{\partial}{\partial z} \\ E_x & 0 & 0 \end{vmatrix} = \hat{y}\frac{1}{\mu_0}\frac{\partial E_x}{\partial z}$$

Interconnection Between Electric and Magnetic Fields of Wave Solutions to Maxwell's Equations

- A general way to solve equations like the one above is to use Laplace transforms.
- In lieu of doing so, we can observe that:
  - The magnetic field is in the y-direction – perpendicular to both the electric field and the direction of propagation.

Summary

- Very far from a source, the source looks like a point and the electromagnetic radiation arising from the source is a spherical wave.
- Locally, a spherical wave looks like a uniform plane wave.
- We can imagine a source of uniform plane waves as an infinite sheet of uniform current.
- Maxwell's equations can be manipulated to provide wave equations describing electromagnetic radiation.

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